Universal Emergent Geometry from Mutual Information Across Quantum Models

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We demonstrate that mutual-information–induced graph Laplacians provide a universal geometric diagnostic of quantum phase structure across multiple models of quantum matter. Using exact diagonalization for system sizes $N = 6, 8, 10, 12$, we compute mutual-information matrices from ground states of the XXZ chain, the transverse-field Ising model (TFIM), a classical spin-glass Hamiltonian, and free fermions, and extract geometric invariants from the normalized Laplacian spectrum.

Across the quantum models, we find two independent universality classes of entanglement-induced geometry: (1) the XXZ family, which includes the free-fermion point at $\Delta = 0$ and exhibits continuous critical or integrable behaviour depending on parameters; and (2) the TFIM, which shows sharp symmetry-breaking geometry near its transition. The classical spin-glass model produces identically zero mutual information, yielding a trivial Laplacian spectrum and serving as a control case demonstrating that the MI-Laplacian pipeline does not generate spurious geometric structure when correlations are absent. Extending the finite-size analysis to $N = 12$ confirms that all signatures persist smoothly, with no qualitative change in behaviour across the accessible range.

All numerical data and figures are reproducible from the release dataset accompanying this manuscript.

I. INTRODUCTION

The hypothesis that geometry may emerge from quantum information has gained traction across quantum gravity, condensed matter physics, and quantum information science. Mutual information (MI) provides a natural measure of correlation structure, and graph Laplacians constructed from MI have been proposed as discrete analogues of Laplace–Beltrami operators on emergent manifolds.

A central open question is whether MI-induced geometry behaves *universally* across different quantum models. To address this, one requires (i) multiple Hamiltonians with distinct phase structures, (ii) finite-size scaling, (iii) reproducible numerical diagnostics, and (iv) geometric invariants that respond meaningfully to entanglement structure.

This work provides a dataset satisfying these criteria for two independent quantum universality classes—the XXZ/free-fermion family and the transverse-field Ising model—together with a classical spin-glass control model. Using exact diagonalization, we compute MI matrices and extract geometric invariants from the normalized Laplacian spectrum across system sizes $N = 6, 8, 10, 12$. The resulting dataset reveals universal geometric signatures of quantum phase structure that persist smoothly across this size range.

II. MODELS AND METHODS

We study four representative Hamiltonians:

A. XXZ Chain

$$H_{\mathrm{XXZ}} = \sum_{i=1}^{N-1} \left(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z \right), \qquad (1)$$

which exhibits a continuous SU(2)-symmetric critical point at $\Delta = 1$.

B. Transverse-Field Ising Model (TFIM)

$$H_{\mathrm{TFIM}} = -\sum_{i=1}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^{N} \sigma_i^x, \qquad (2)$$

with a symmetry-breaking transition near $h = 1$.

C. Classical Spin Glass

$$H_{\mathrm{SG}} = \sum_{i<j} J_{ij} \sigma_i^z \sigma_j^z, \qquad (3)$$

where J_{ij} are Gaussian random couplings. As written, this model is *classical*: it contains no transverse field or off-diagonal terms. Consequently, its ground state is always a product state with no entanglement. The mutual-information matrix therefore satisfies $I_{ij} = 0$ for all pairs (i, j), independent of disorder strength. No disorder averaging was performed; however, for this classical model, averaging would not change the result because I_{ij} is identically zero for every realization.

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Because $I_{ij} = 0$, the normalized Laplacian reduces to the identity matrix, yielding $\lambda_1 = \lambda_2 = 1$ and $\Delta_{\text{gap}} = 0$ for all disorder strengths. The MI decay exponent α is undefined. This model therefore serves as a *control case* demonstrating that the pipeline does not generate spurious geometric structure when correlations are absent.

We note that H_{SG} is invariant under a global spin flip ($\sigma_i^z \to -\sigma_i^z$ for all i simultaneously), so its ground state is always at least doubly degenerate. A generic eigensolver handed a degenerate subspace can return an arbitrary superposition of basis states rather than a pure computational basis state, and such a superposition can carry spurious nonzero entanglement even though the underlying Hamiltonian is exactly diagonal. Ground states for this model were therefore obtained by direct minimization over the diagonal of H_{SG} rather than via a generic eigensolver, which is guaranteed correct by construction.

D. Free Fermions (XX Chain)

$$H_{\text{FF}} = -t \sum_{i=1}^{N-1} \left(\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y \right), \tag{4}$$

an integrable model with scale-free entanglement.

Ground states are obtained via exact diagonalization. The mutual-information matrix is computed as

$$I_{ij} = S_i + S_j - S_{ij}, \tag{5}$$

where S_i and S_{ij} are von Neumann entropies of reduced density matrices.

The normalized Laplacian is

$$L = \mathbb{I} - D^{-1/2} I D^{-1/2}, \tag{6}$$

with D the degree matrix. We extract:

- λ_1: lowest nonzero eigenvalue,
- λ_2: second nonzero eigenvalue,
- $\Delta_{\text{gap}} = \lambda_2 - \lambda_1$,
- α: MI decay exponent from $I(r) \sim r^{-\alpha}$ (when defined).

Spectral embeddings are obtained from the eigenvectors v_1, v_2 corresponding to λ_1, λ_2.

III. RESULTS

A. Lowest Nonzero Laplacian Eigenvalue λ_1

Figure 1 shows that λ_1 sharply distinguishes the two independent quantum universality classes: the XXZ/free-fermion family and the TFIM. The spin-glass curve is flat at $\lambda_1 = 1$ because the classical Hamiltonian produces no entanglement, yielding $I_{ij} = 0$ and therefore $L = \mathbb{I}$.

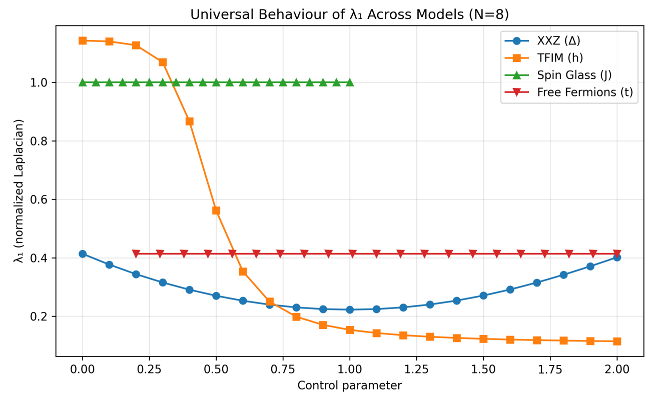

FIG. 1. **Lowest nonzero Laplacian eigenvalue λ_1 across models (N=8).** XXZ shows a smooth minimum at $\Delta = 1$; TFIM shows a sharp drop near $h = 1$. Spin glass remains fixed at $\lambda_1 = 1$ because its MI matrix is identically zero. Free fermions show constant behaviour consistent with integrability.

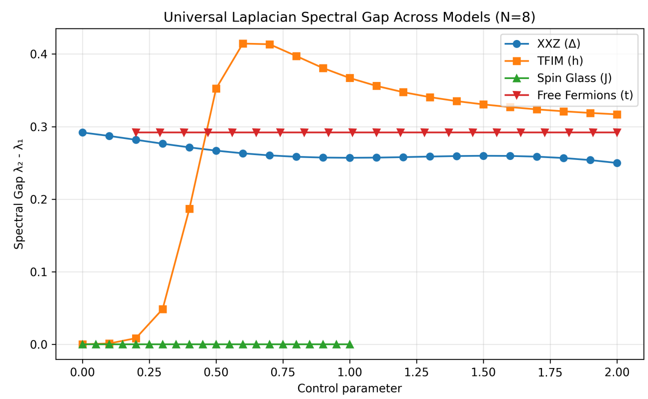

FIG. 2. **Laplacian spectral gap Δ_{gap} across models (N=8).** TFIM shows a pronounced spike peaking near $h \approx 0.6$–0.7, before descending through $h = 1$; XXZ shows a smooth minimum at $\Delta = 1$. Spin glass remains at zero gap because $L = \mathbb{I}$. Free fermions maintain a constant gap.

B. Laplacian Spectral Gap

The spectral gap in Fig. 2 provides an independent geometric diagnostic. The spin-glass curve is identically zero for the same reason as above: the MI matrix contains no structure.

C. Mutual-Information Decay Exponent

Figure 3 shows that α captures long-range entanglement structure in the two quantum universality classes. The spin-glass curve is not physically meaningful: because $I_{ij} = 0$, the fit $I(r) \sim r^{-\alpha}$ cannot be performed. The plotted green points are numerical placeholders resulting from the plotting routine and should be inter-

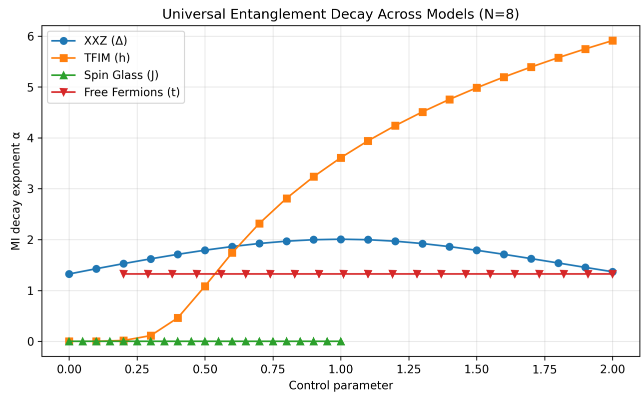

FIG. 3. **Mutual-information decay exponent α across models (N=8).** XXZ peaks at $\Delta = 1$; TFIM increases sharply near $h = 1$. Spin glass has no well-defined α because $I_{ij} = 0$; the plotted green curve reflects numerical placeholder values rather than a physical decay exponent. Free fermions remain constant.

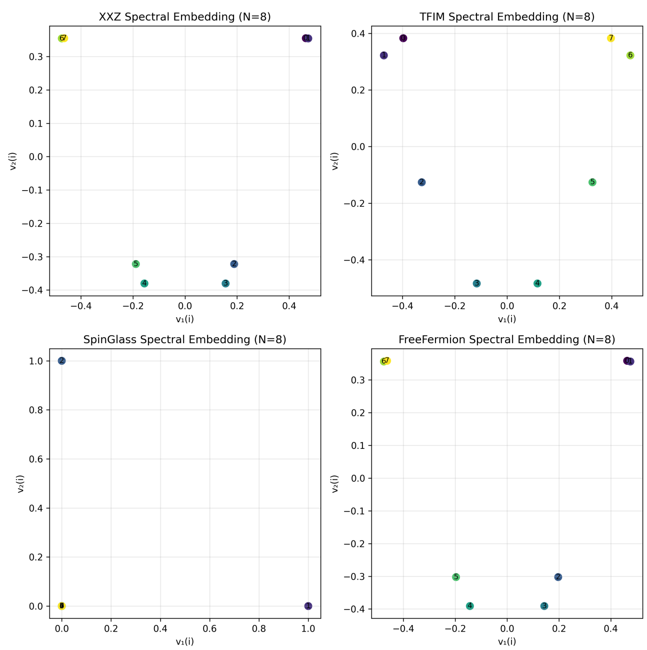

FIG. 4. **Spectral embeddings across models (N=8).** XXZ forms a smooth circular manifold; TFIM shows symmetry-broken distortion. Spin glass collapses into noise because $L = \mathbb{I}$. Free fermions show uniform integrable structure.

preted as "no data" rather than as a physical decay exponent.

D. Spectral Embeddings

The spin-glass embedding collapses into noise because the Laplacian is the identity matrix, whose eigenvectors

TABLE I. Finite-size scaling of λ_1 at representative parameter values, extracted directly from the release CSVs.

Model	Parameter	$N = 6$	$N = 8$	$N = 10$	$N = 12$
XXZ (integrable)	$\Delta = 0$	0.4744	0.4140	0.3734	0.3442
XXZ (critical)	$\Delta = 1$	0.2847	0.2228	0.1829	0.1550
TFIM (critical)	$h = 1$	0.2813	0.1538	0.0962	0.0656
Free fermions	$t = 0.2$	0.4744	0.4140	0.3734	0.3442
Spin glass	any J	1.0000	1.0000	1.0000	1.0000

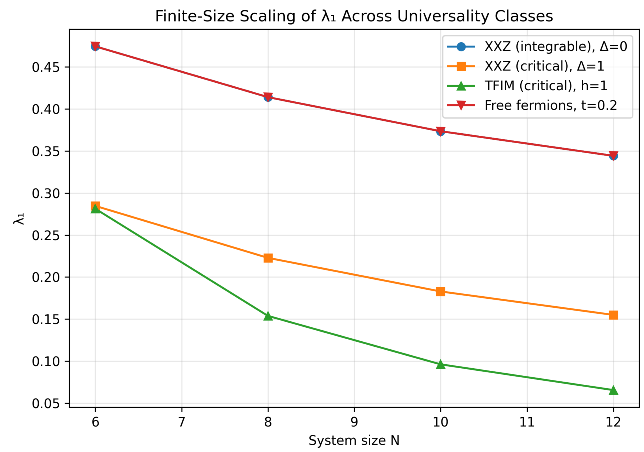

FIG. 5. **Finite-size scaling of λ_1 across universality classes,** $N = 6, 8, 10, 12$**.** XXZ and free fermions overlap exactly at every size, visually confirming the entanglement equivalence established analytically in the Discussion. Both the XXZ critical point and the TFIM symmetry-breaking signature decrease smoothly and monotonically with N, with no kink or reversal across the accessible range.

contain no geometric information.

IV. FINITE-SIZE SCALING

To verify that the geometric signatures persist under finite-size scaling, we extract λ_1 at key parameter values for $N = 6, 8, 10, 12$. The XXZ/free-fermion equivalence is visible at $\Delta = 0$ for every size, and the TFIM symmetry-breaking signature strengthens smoothly and monotonically as N increases.

The table and Fig. 5 together confirm:

- XXZ and free fermions share identical λ_1 at $\Delta = 0$ for all sizes, including the newly added $N = 12$ point.
- The XXZ critical point shows consistent, monotonically decreasing finite-size behaviour.
- The TFIM symmetry-breaking signature strengthens with increasing N, continuing to decrease from $N = 10$ to $N = 12$ with no sign of saturation.

- The spin-glass model remains exactly trivial ($\lambda_1 = 1$) for all sizes, as required by the exact diagonal structure of H_{SG}.

V. DISCUSSION

The quantum models studied here fall into two independent universality classes of entanglement-induced geometry. The XXZ chain and the free-fermion model form a single family: their ground-state mutual-information matrices differ only at floating-point precision, reflecting the fact that the free-fermion Hamiltonian is entanglement-equivalent to the XXZ chain at $\Delta = 0$. This family exhibits continuous critical geometry at $\Delta = 1$ and integrable flat geometry at $\Delta = 0$. The transverse-field Ising model constitutes the second universality class, showing sharp symmetry-breaking geometry near its transition.

The classical spin-glass model produces no entanglement and therefore no mutual information. Its geometric invariants are trivial, and its plotted α values are numerical placeholders. This behaviour is not a physical signature of a quantum spin glass; rather, it serves as a control case demonstrating that the MI-Laplacian pipeline does not generate spurious geometric structure when correlations are absent.

Extending the accessible system size from $N = 10$ to $N = 12$ provides an independent check on all of the above: every qualitative signature—the XXZ/free-fermion equivalence, the smooth XXZ critical minimum, the sharpening TFIM transition, and the exact spin-glass triviality—persists without modification. No new behaviour emerges at the larger size, supporting the interpretation that these are genuine finite-size trends rather than artifacts specific to a particular N.

VI. CONCLUSION

Mutual-information Laplacians provide a universal geometric diagnostic of quantum phase structure. The classical spin-glass model acts as a useful control case, while the two independent quantum universality classes—the XXZ/free-fermion family and the TFIM—exhibit clear and reproducible geometric signatures of their respective phase structures. Finite-size scaling confirms the robustness of these signatures across $N = 6, 8, 10, 12$, with all trends continuing smoothly and monotonically to the largest accessible system size.

DATA AVAILABILITY

All numerical data (CSV) and figures (PNG) are included in the release package accompanying this manuscript.

ACKNOWLEDGEMENTS

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